01 · Is there a Phillips-curve trade-off in Brazil? (2012–2026)

This report's text is generated automatically from the code's outputs. The main output is the quantitative part: data, checks, models and tables. It extracts the quantitative work of a longer paper I wrote for a UFRJ econometrics course; the full paper is available on request.

Question

Does higher unemployment go with lower inflation in Brazil, once inflation inertia and expectations are in the model?

Result

• In the static model the unemployment coefficient is -0.002 (SE 0.010), with no stars: no static trade-off.

• In the dynamic model, lagged inflation has a coefficient of 0.398 and the Focus expectation 0.686, both significant at 1%; unemployment stays insignificant at 0.007.

• R^2 rises from 0.000 to 0.421, on 169 monthly observations.

• H0: $\gamma + \delta = 1$ is not rejected (p = 0.432): the data do not reject a vertical long-run Phillips curve.

• With Newey–West errors the standard error on the expectation term rises from 0.111 to 0.177; it stays significant.

• With seasonal dummies the expectation coefficient is 0.783 and R^2 0.523; without the COVID months unemployment is 0.001.

Reading: Brazilian inflation is driven by inertia and expectations; labour-market slack barely shows up. Descriptive, not causal: expectations are endogenous and the model uses the unemployment rate, not the gap to its natural rate.

Data checks

17 pass, 1 warnings, 0 fail (details in output/data_checks.json).

Main tables

Curva de Phillips: especificação estática vs. dinâmica (MQO)

	Dependent variable: IPCA mensal	
	Estática	Dinâmica
	(1)	(2)
Desemprego u_t	-0.002	0.007
	(0.010)	(0.008)
Inflação defasada π_t-1		0.398***
		(0.063)
Expectativa E_t[π_t+1]		0.686***
		(0.111)
Intercept	0.491***	-0.066
	(0.102)	(0.097)
Durbin–Watson	0.927	2.558
Efeito de longo prazo β/(1-γ)	—	0.0114
Breusch–Godfrey(12) p	<0.001	7.59e-06
H_0: γ+δ=1 (p-valor)	—	0.432
Observations	170	169
R^2	0.000	0.421
Adjusted R^2	-0.006	0.411
Residual Std. Error	0.372 (df=168)	0.286 (df=165)
F Statistic	0.055 (df=1; 168)	40.014*** (df=3; 165)

Source: output/q2_tab_phillips_dinamica.tex

Modelo dinâmico: erros-padrão MQO usual vs. HAC (Newey–West)

	Dependent variable: IPCA mensal	
	MQO usual	MQO + HAC (L=4)
	(1)	(2)
Desemprego u_t	0.007	0.007
	(0.008)	(0.007)
Inflação defasada π_t-1	0.398***	0.398***
	(0.063)	(0.051)
Expectativa E_t[π_t+1]	0.686***	0.686***
	(0.111)	(0.177)
Intercept	-0.066	-0.066
	(0.097)	(0.111)
Observations	169	169
R^2	0.421	0.421
Adjusted R^2	0.411	0.411
Residual Std. Error	0.286 (df=165)	0.286 (df=165)

Source: output/q2_tab_hac_comparacao.tex

Robustez do modelo dinâmico (coeficientes-chave)

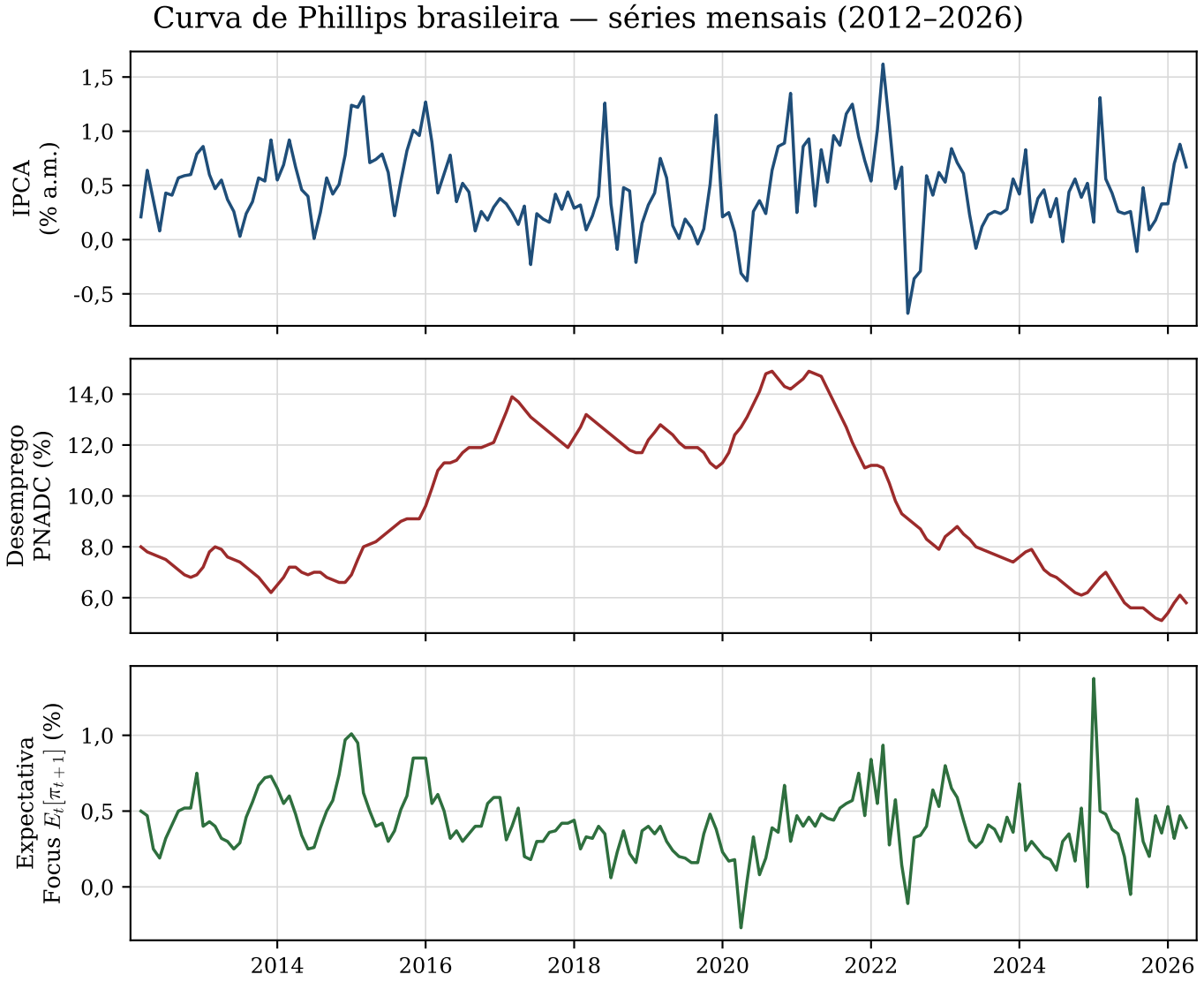
	Dependent variable: IPCA mensal			
	Base	+ Sazonal	Sem COVID	+ Tendência
	(1)	(2)	(3)	(4)
Desemprego u_t	0.007	0.008	0.001	0.007
	(0.008)	(0.007)	(0.009)	(0.008)
Inflação defasada π_t-1	0.398***	0.394***	0.420***	0.398***
	(0.063)	(0.064)	(0.066)	(0.063)
Expectativa E_t[π_t+1]	0.686***	0.783***	0.649***	0.690***
	(0.111)	(0.125)	(0.113)	(0.113)
Tendência linear				0.000
				(0.000)
Dummies sazonais	Não	Sim	Não	Não
Breusch–Godfrey(12) p	7.6e-06	0.00055	4.4e-05	4.5e-06
Observations	169	169	153	169
R^2	0.421	0.523	0.423	0.421
Adjusted R^2	0.411	0.480	0.411	0.407
Residual Std. Error	0.286 (df=165)	0.268 (df=154)	0.278 (df=149)	0.286 (df=164)
F Statistic	40.014*** (df=3; 165)	12.068*** (df=14; 154)	36.427*** (df=3; 149)	29.851*** (df=4; 164)

Source: output/q2_tab_robustez.tex

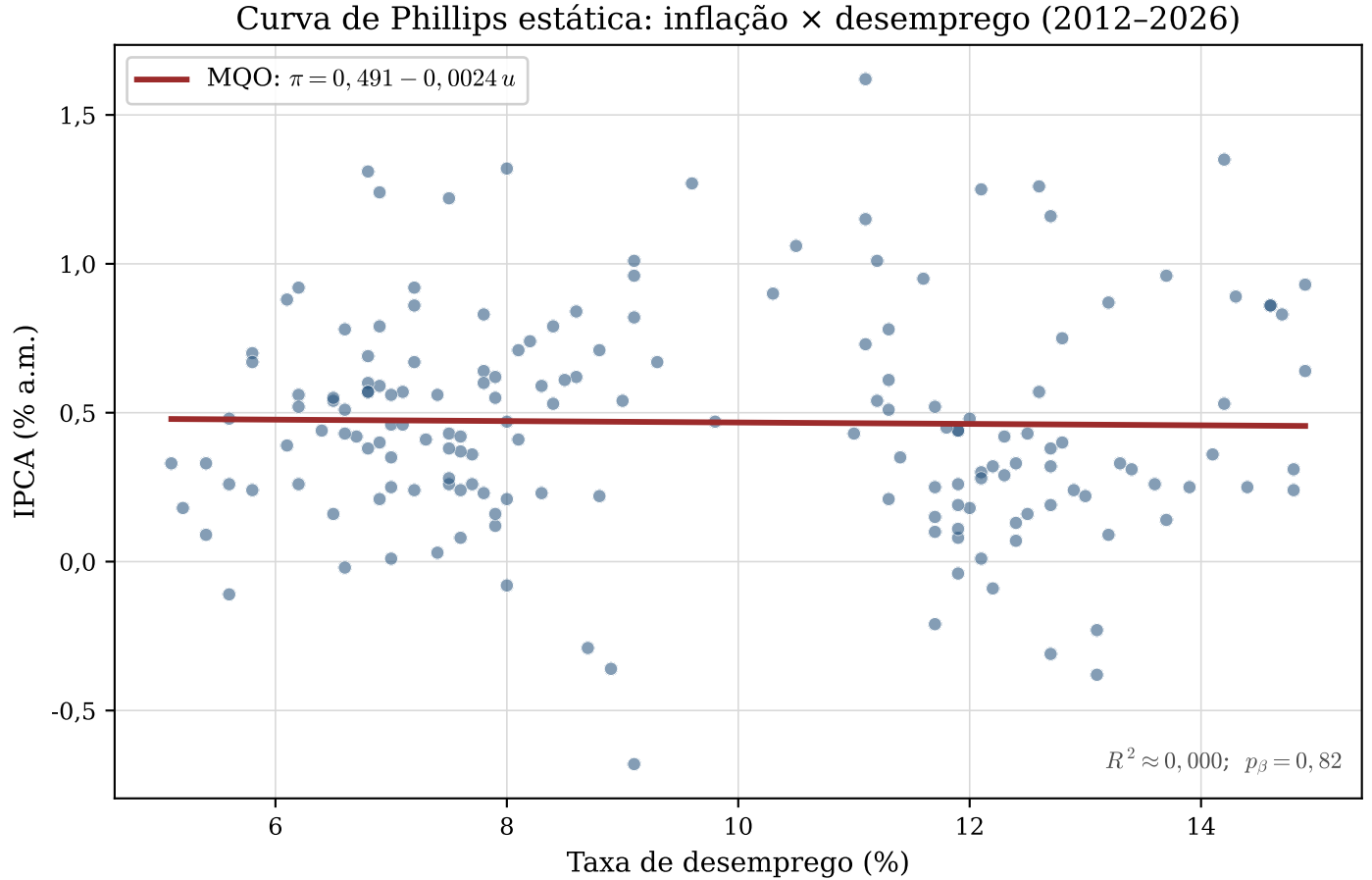
Estacionariedade: ADF (H_0: raiz unitária) e KPSS (H_0: estacionária)

	ADF	KPSS			
(Ir)2-3 (Ir)4-5 Série	Estat.	p	Estat.	p	Conclusão (5%)
IPCA (% a.m.)	-7.12	0.000	0.13	>0.10	Estacionária
Desemprego (%)	-1.35	0.607	0.40	0.076	Persistente (evid. mista)
Expectativa E_t[π_t+1] (%)	-1.73	0.418	0.34	>0.10	Persistente (evid. mista)
Em primeira diferença (apenas ADF):					
IPCA (% a.m.)	-8.10	0.000	—	—	Estacionária
Desemprego (%)	-2.63	0.087	—	—	Estac. a 10%

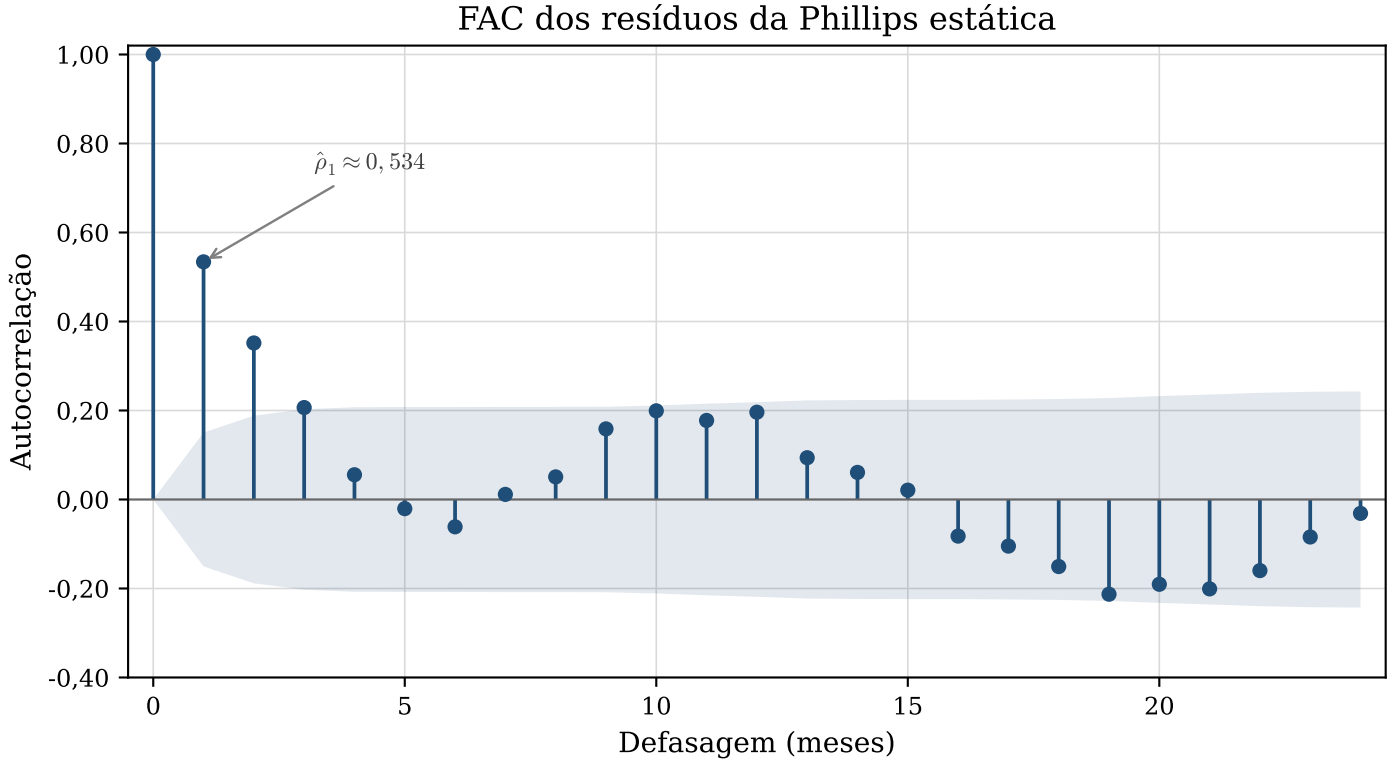
Source: output/q2_tab_estacionariedade.tex



Fonte: BCB — SGS (IPCA 433; desocupação PNADC 24369) e Expectativas/Focus. Elaboração própria.



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Elaboração própria a partir dos resíduos da regressão estática.